08A — Deep-Dive Feature Audit: Outside-Sect World & Ecology (Part 1 of 6)

Macro Geopolitics, 19 Rival Sects, Foreign Cognition Tiers, Diplomacy & Treaties, Vassal Relief, 6 Contestable Resource Nodes & Sect Siege Warfare

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Autonomous Xianxia Sandbox**. Rival sects on the continental map must breathe under the exact same physical, economic, and 43-Path rules as the player's sect: armies eat real rations, spies risk capture, vassal rescue requires marching a real relief force, and resource harvests obey strict calendar and Closed-Loop conservation laws!
- **Master Roadmap for Category 08 (`08A–08F`)**:
  - **`08A` (This Document)**: **Macro Geopolitics, 19 Rival Sects, Diplomacy/Treaties, Vassal Tributes & Distress, 6 Resource Nodes & Sect Siege Warfare (`expedition.js`, `content.js`, `sect.js`)**
  - `08B`: Expeditions, Caravans, Midnight Black Market, Tavern Bounties, Cross-Sect Exchange/Espionage & Living Chronicle (`v2.6.0`)
  - `08C`: Secret Realms, 43-Floor Nine-Dao Pagoda, 5-Elemental Realms, 4 Sanctums, 5-Stage Branching Nodes, 4 Micro-Locations & Immortal Tombs
  - `08D`: Wilderness Ecology, Roaming Spirit-Beast Packs, Taming/Pets/Breeding/Menagerie/Symbiosis & Wild Resource Gathering
  - `08E`: Southern Spirit River Fishing, Spirit Cuisine, Underwater Chests, Wandering Figures, Visitors & Legendary Clue Quests
  - `08F`: Grand Finale — Holistic Overview of the Entire Outside-Sect World & Ecology (`W.world.resourcePool` Equilibrium, 200-Year Scaling & Master `P0->P3` Roadmap)

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0. Status of Prior `v2.4.0` Outside-Sect QA Audit (`Filtered Baseline`)

Per `JS2_V2_4_0_FIX_TRACKER_TH.md`, `OUT-01, 02, 03, 04, 06, 09, 10, 11, 13` were patched in `v2.4.0`. This document focuses strictly on **unfixed live defects (`OUT-08,`**

12, 15, 16, 17, 19) plus **newly uncovered deep architectural & emergent-AI flaws** in JS2-v2.6.0.

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1. [GEO-RIVAL-01] 12-Path Fossilization of the 19 Rival Sects & Infinite PILLAGE / Tribute Treasury Vacuum!

1. **Zero Native Representation of the 31 Grand Dao Paths Across Rival Sects (W.world.sects)**:
   - While player disciples awaken across **43 Major Paths**, the 19 rival sects on the continental map remain hardcoded to legacy 12-path templates! The Jianghu lacks a Bushido Clan (SHADOW_SAMURAI), Shinobi Village (NINPO_SHADOW), Dharma Monastery (BUDDHA_DHARMA), Astral Observatory (ASTRO_DIVINATION), Mercantile Consortium (MERCANTILE_DAO), or Occult Sanctum (OCCULT_MAGE).
   - **Fix**: Enrich rival sect secondary/branch lineages (secondaryPaths & champion weapon loadouts) across the 31 Grand Dao Paths—so cross-sect exchanges (dispatchExchangeDisciple), tournament guests, and prisoner interrogations bring rare Grand Dao arts into circulation organically!
2. **Unbounded PILLAGE & Vassal Tribute Extraction (expedition.js:2721, 2835)**:
   - v330 (COMPENSATE_STONES) added rivalSect.treasury, **yet applySiegeResolution(PILLAGE) and collectVassalTributes still mint stones by formula rather than bounding extraction by rivalSect.treasury** (and OUT-19: at W.day === 0, curDay > 0 skips the duplicate-collection guard!).
   - **Fix**: Give every rival sect a Closed-Loop sect.treasury funded monthly from W.world.resourcePool.stones according to its tier and owned RESOURCE_NODE_DEFS. Cap PILLAGE (max 45% of targetSect.treasury) and collectVassalTributes (strict integer periodId = Math.floor(W.day / 90) fixing OUT-19, capped by vassal.treasury)!

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2. [GEO-WAR-02] Unfixed Siege, Vassal-Relief & Resource-Node Exploits (OUT-08, 12, 15, 16, 17)!

1. **Zero-Ration Sieges & Misleading Deterministic winProb (OUT-17, js/expedition.js:2468–2700)**:
   - Ordinary expeditions (EXPED.depart) consume real meal rations, **whereas marching an army to siege a rival sect (departSectSiege) sets ex.supplies = 0, consumes 0 meals, inflicts 0 real HP/injury attrition on**

victorious attackers, and displays a `winProb` percentage while resolving deterministically via `attack >= defense * coeff`!
    - **Fix**: Deduct real expedition `meal` rations (`scaled by party size * travelDays`, modified by `EFX_SUPPLY_EFFICIENCY 05B` and `CAMP_SUPPLY` outposts!), resolve siege phases with seeded `W.rng` rolls around `winProb` (or honest tactical combat simulation), and apply proportional HP loss, durability wear, and injury rolls based on enemy defense resistance!
2. **Instant Zero-Party Vassal Relief (`OUT-08`, `js/expedition.js:2903–2927`)**:
    - Clicking `REINFORCE` on a Vassal Distress Call requires **no party, `0` rations, `0` travel time, and `0` combat**, immediately granting `loyalty +20, attitude +15, fame +4`!
    - **Fix**: Require dispatching a real available relief party (`>= 2` cultivators, `!away, !jail`) that travels to the vassal before `expiresDay` and fights the besieging threat!
3. **Premature Seasonal Node Harvest (`OUT-12`, `expedition.js:3153–3184`) & Repeat `claimResourceNode` Fame Farming (`OUT-15`, `line 3026`)**:
    - **Bug A (`OUT-12`)**: UI passes `{ seasonal: true }` unconditionally into `harvestResourceNodes`, letting a newly captured node cash out a **90-day seasonal payout on Day 361** (`not a seasonal boundary!`) while sharing a single daily guard variable that blocks normal daily yield! Split `lastDailyHarvestDay` from `lastSeasonalPeriod = Math.floor(W.day / 90)` and require holding the node for ≥ 15 days before claiming seasonal yield!
    - **Bug B (`OUT-15`)**: Guard `claimResourceNode` directly at the engine layer: reject if `rn.ownerSectId === 'PLAYER'` or if lacking a valid arrival expedition receipt!
4. **Late-Arrival Free Loot on Contested World Anomalies (`OUT-16`, `expedition.js:3313–3364`)**:
    - `arrivedBeforeRival || partyPower >= 0.85 * rivalPower` lets a slow party arrive *after* `rivalEtaDay` and still loot the treasure with zero battle! When `W.day >= rivalEtaDay`, mark the anomaly claimed by the rival party so latecomers must either **fight the rival's expedition party (`attitude -18`, injury risk)** or return empty-handed!

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3. `[GEO-AUTON-03]` Broken `'social.diplomacy' in {}` Typo & Disconnected Geopolitical Autonomy (`commander` / `envoy` / `spymaster`)!

1. **Dead Ternary Typo (`js/expedition.js:44`)**:

- `'social.diplomacy' in {} ? HC.eff(n, 'social.diplomacy') : ...` evaluates `'social.diplomacy' in {}` (`always false!`), completely ignoring NPCs' actual `social.diplomacy` trait across diplomatic calculations! Replace with `HC.eff(n, 'social.diplomacy')`.

2. **Zero Autonomous Frontier Expansion, Vassal Rescue, or Allied Mutual-Defense**:
   - Wire **`commander` + `HAWK_MARTIAL` Master**: When `W.sect.stones` and `meal` reserves are healthy and an adjacent `RESOURCE_NODE` is unowned (or held by a hated `RIVAL_FEUD` enemy weaker than `0.75×` our strike force), autonomously organize a `departResourceClaim` expedition!
   - Wire **`envoy`**: Autonomously dispatch relief parties to `distressCall` vassals (`when generosity/empathy >= 48`) and upgrade `'TRADE_CARAVAN_ACCORD'` $\to$ `'ALLIANCE'` when `attitude >= 50`!
   - Wire **Mutual-Defense Alliances (`treaty === 'ALLIANCE'`)**: When a hostile sect launches a major Raid against our sect (`EXPED.raidTick`), give each active `'ALLIANCE'` partner (`attitude >= 55`) a chance to send 1–2 allied guest champions to fight alongside our defenders at the `gate`!

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4. Mandatory Verification Suite Additions (`08A`)

1. **Seasonal vs Daily Node Harvest Isolation Test (`OUT-12` & `OUT-15`)**: Capture a node on Day 361; assert `{ seasonal: true }` on Day 361 yields `0` seasonal payout without blocking daily yield, and repeat `claimResourceNode` on a `PLAYER`-owned node returns `{ ok: false, reason: 'already-owned' }` with `0` fame gain.
2. **Real Relief Party & Siege Supply Test (`OUT-08` & `OUT-17`)**: Assert `respondVassalDistress('REINFORCE')` without an available party fails, and `departSectSiege` deducts real `meal` rations.
3. **Closed-Loop Rival Sect Treasury Test**: Assert `PILLAGE` and `collectVassalTributes` never extract more Spirit Stones than `targetSect.treasury` holds.